Lecture 4 - Evaluation

ECE 447: Data Analysis and Machine Learning for Engineers

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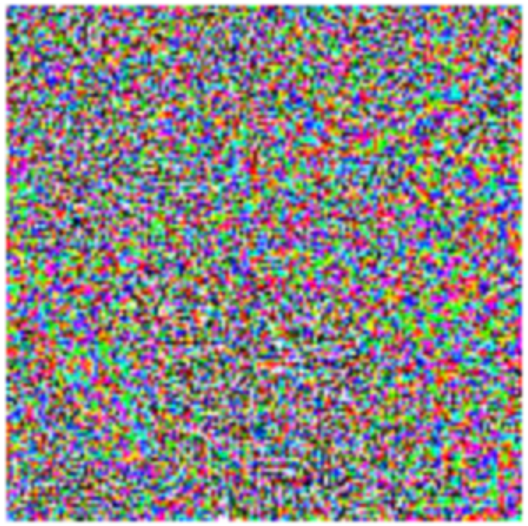
Adversarial Attack

This slide is informational, excluded from assessment.

$+ \quad 0.007 \times$



$=$

A smart, optimized perturbation to attack the ML system!

Human "Panda" 99.99 %

An ML Classifier "Panda" 57.7 %

"Panda" 99.99 %

"Gibbon" 99.3 %

https://www.tensorflow.org/tutorials/generative/adversarial_fgsm

This Lecture Covers ...

- Prediction Scores (two-class)
  - ROC curve.
  - Area under the curve (ROC index).
  - Gain and lift.

- Performance Measure
  - Multinomial targets.
  - Continuous targets.

— *ML for Predictive Data Analytics*

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Prediction Scores

Many classification models provide not *target value* but a *prediction score*

0 or 1

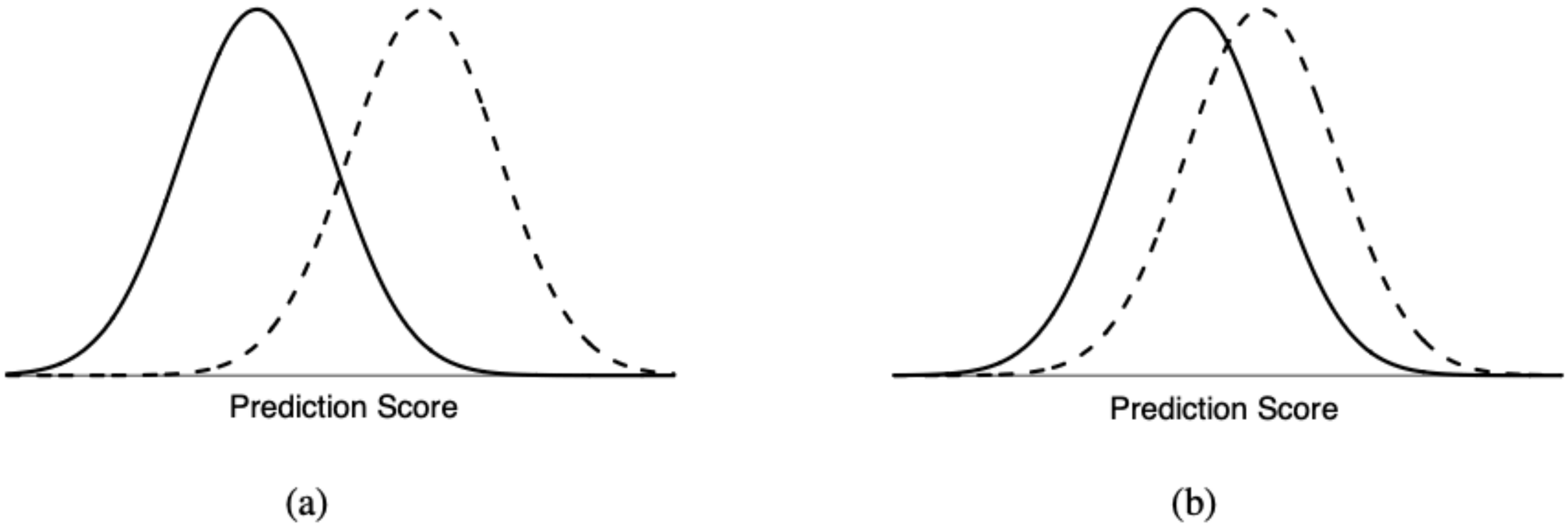
Value in a range

Hence, we use a threshold to map the score into an estimated target value.

$$\text{threshold}(\text{score}, 0.5) = \begin{cases} 1 & \text{if score } \geq 0.5 \\ 0, & \text{otherwise} \end{cases}$$

Prediction Scores (Idealistic Example)

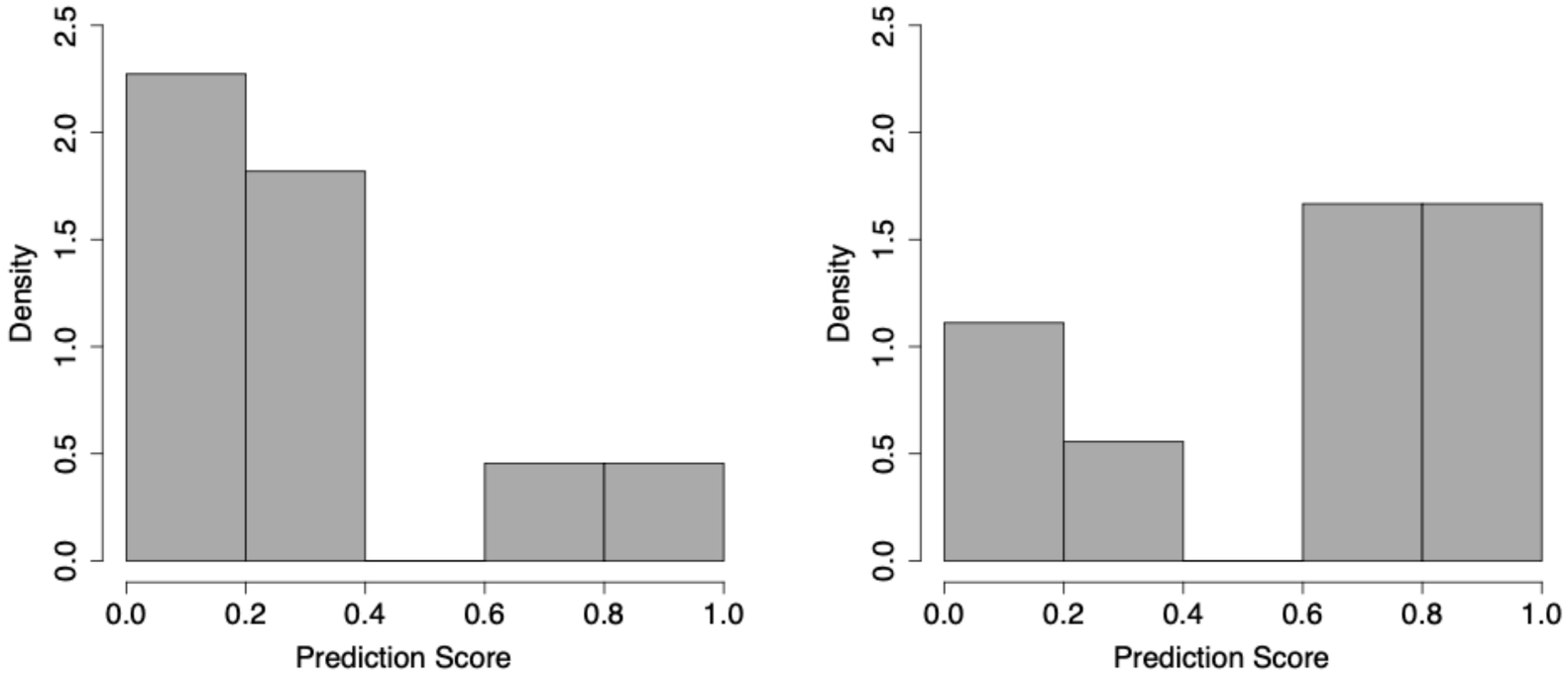
Here is a distribution of scores from the two classes (solid and dashed curves). Which threshold you suggest for each scenario?



Imagine these are probability density functions.

figure from ML for Predictive Data Analytics textbook.

Prediction Scores from the Email Example (Realistic)



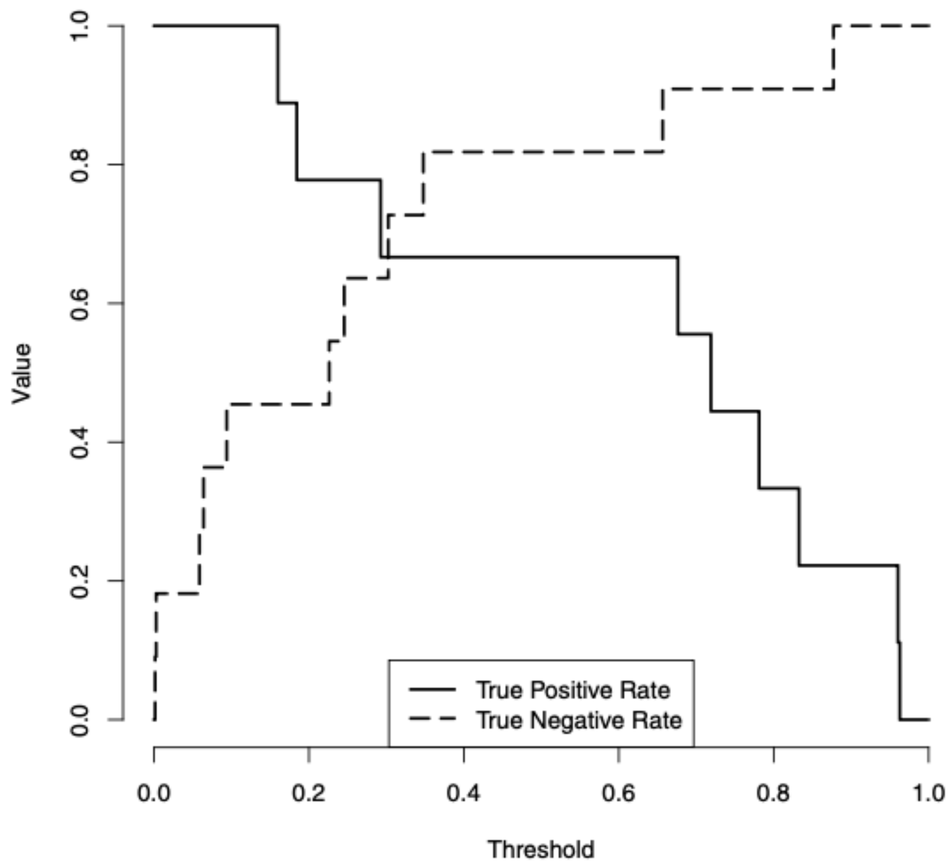
(a) *spam* (b) *ham*

What to do here? Not so much clear (we discuss solutions in this lecture).

figure from ML for Predictive Data Analytics textbook.

How to Choose the Threshold on Prediction Scores?

The threshold can be controlled, hence control accordingly, TPR and TNR.



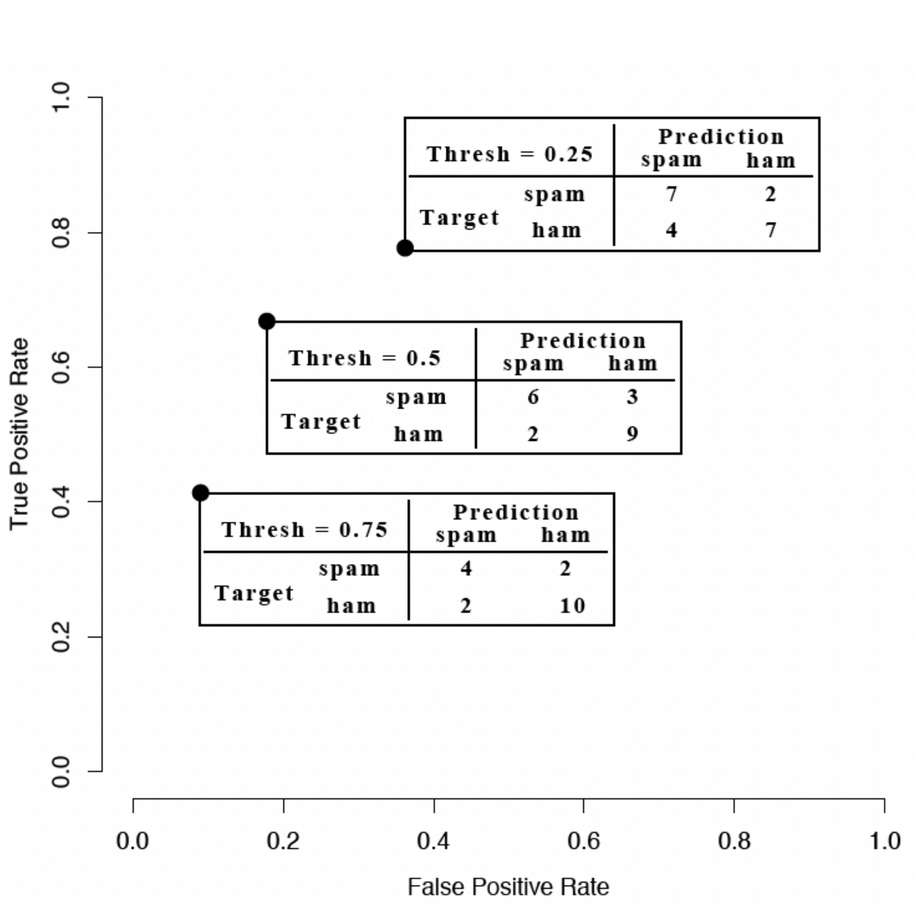
For every value of threshold, we have different values of TPR and TNR.

There a trade-off between accuracy for predictions or positive target levels and accuracy for prediction of negative target levels.

figure from ML for Predictive Data Analytics textbook.

Receiver Operating Characteristic (ROC) Space

Draw the relation between TPR and TNR for every feasible threshold value.



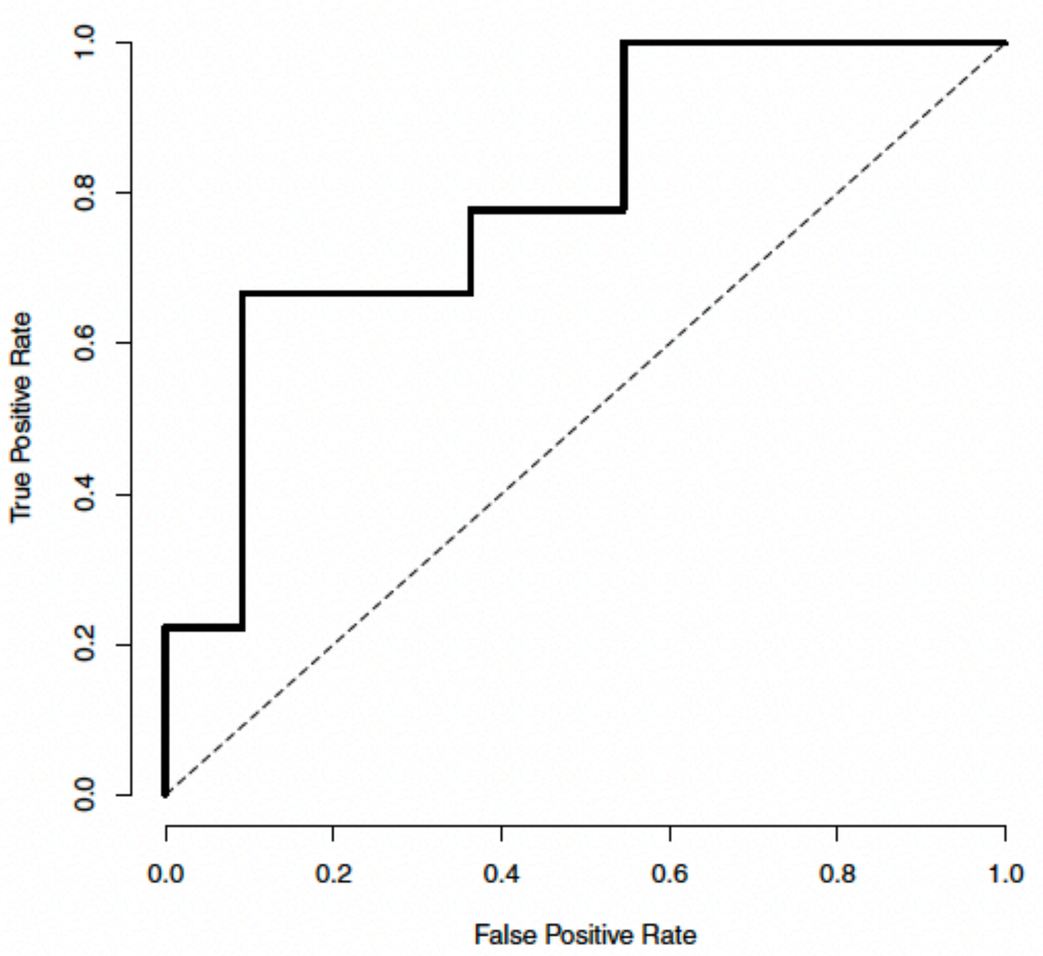
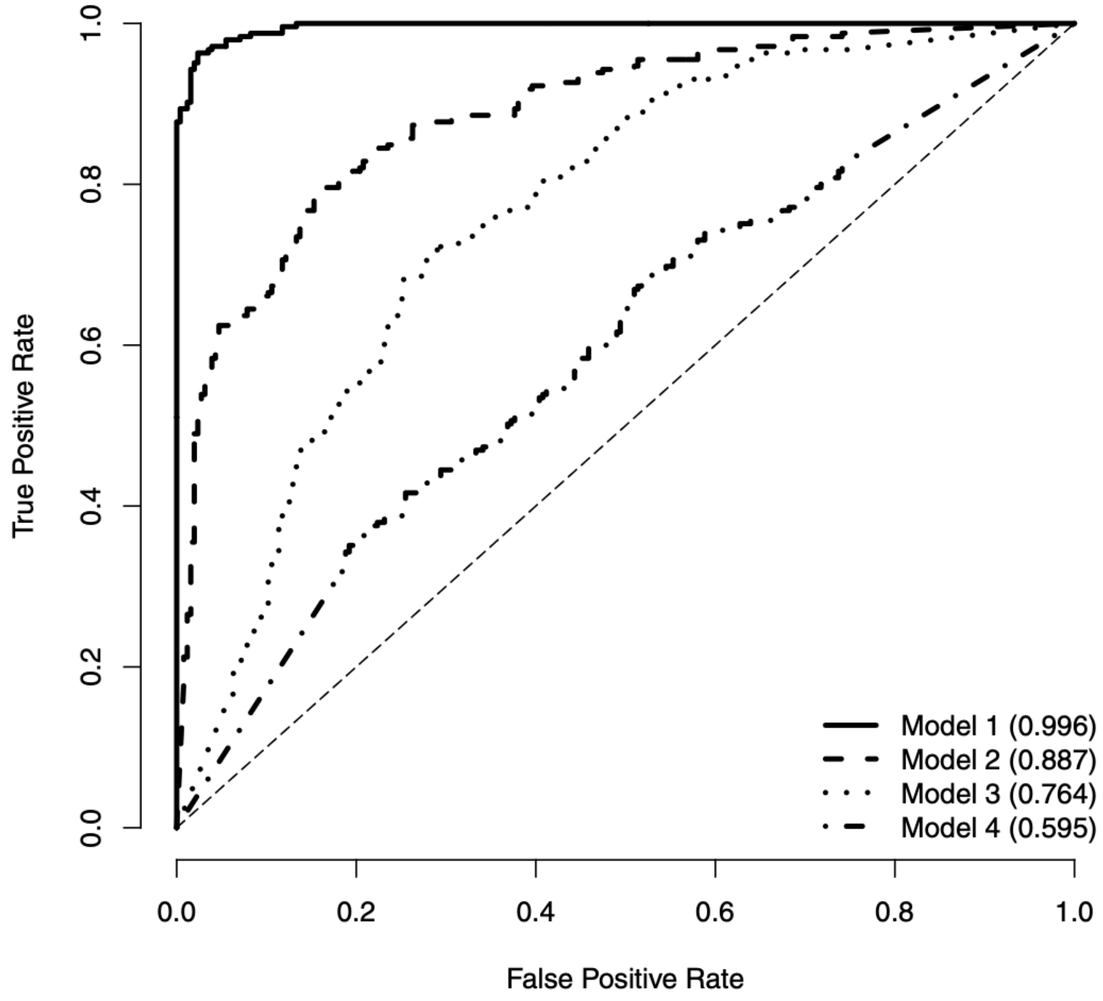


figure from ML for Predictive Data Analytics textbook.

Model Comparison Using ROC Curves (Visualized)



Which model you should choose (performing better)?

figure from ML for Predictive Data Analytics textbook.

Quantify ROC with Area Under the Curve (AUC)

Area Under the Curve (AUC) or ROC Index

It measures the area underneath an ROC curve.

Good model will appear at the very top left-hand corner of ROC. Hence, better the model, higher the AUC. The ROC index is

$$\sum_{i=2}^{|\mathbf{T}|} \frac{(\text{FPR}(\mathbf{T}[i]) - \text{FPR}(\mathbf{T}[i-1])) \times (\text{TPR}(\mathbf{T}[i]) + \text{TPR}(\mathbf{T}[i-1]))}{2}$$

where $\mathbf{T}$ is a set of thresholds, and $|\mathbf{T}|$ is the number of thresholds tested.

Intuitions on ROC

There are no hard and fast rules about what constitutes an acceptable value for the ROC index.

A "good" value of ROC is application specific.

Rule of thumb: ROC index above 0.7, then a *strong* model.
ROC index below 0.6, then a *weak* model.

The ROC index is quite robust to imbalanced data.

Common choice for practitioners to compare multiple modelling techniques to one another.

ROC Probabilistic Interpretation and Gini Index

- AUC of 0.5 is random chance.
- AUC of 1 is perfect.

ROC Index

The probability that a model will assign a higher rank to a randomly selected positive instance than to a randomly selected negative instance.

Gini Index

$$\text{Gini coefficient} = (2 \times \text{ROC index}) - 1$$

In the range $[0,1]$

Commonly used in financial modelling; you might hear about it in credit scoring scenarios.

How to Study the Impact of Model on one Target Value (Level)?

Goal: Want to know

how well the model is making prediction for specific instances (data examples);

rather than how the model distinguishes between two target values (levels), which looks into overall population statistics.

We can use performance measure of **gain** and **lift**.

Intuition on What Gain and Lift Quantify

We do this to quantify gain and lift

Consider the following

- If we were to rank the instances in a test set in descending order of the prediction scores assigned to them by a well-performing model,
- we would expect the majority of the positive instances to be toward the top of this ranking.
- The **gain** and **lift** measures attempt to quantify to what extent a set of predictions made by a model meet this assumption.

Gain of Prediction Scores

The ground-truth
known target value

Gain

A measure of how many of the positive instances in the overall test set are found in a particular decile (***dec***).

$$\text{gain}(\boldsymbol{dec}) = \frac{\text{number of positive test instances in decile } \boldsymbol{dec}}{\text{number of positve test instances}}$$

Cumulative Gain

$$\text{cumulative gain}(\boldsymbol{dec}) = \frac{\text{number of positive test instances in all decile up to } \boldsymbol{dec}}{\text{number of positve test instances}}$$

Deciles and Ranked Model Predictions and Scores from Email Classification Example

Decile	ID	Target	Prediction	Score	Outcome
1st	9	spam	spam	0.960	TP
	4	spam	spam	0.963	TP
2nd	18	spam	spam	0.833	TP
	20	ham	spam	0.877	FP
3rd	6	spam	spam	0.719	TP
	10	spam	spam	0.781	TP
4th	17	ham	spam	0.657	FP
	8	spam	spam	0.676	TP
5th	5	ham	ham	0.302	TN
	14	ham	ham	0.348	TN
6th	16	ham	ham	0.246	TN
	1	spam	ham	0.293	FN
7th	2	spam	ham	0.184	FN
	3	ham	ham	0.226	TN
8th	19	ham	ham	0.094	TN
	12	spam	ham	0.160	FN
9th	15	ham	ham	0.059	TN
	13	ham	ham	0.064	TN
10th	7	ham	ham	0.001	TN
	11	ham	ham	0.003	TN

Gain and Cumulative Gain for Email Example

Decile	ID	Target	Prediction	Score	Outcome
1st	9	spam	spam	0.960	TP
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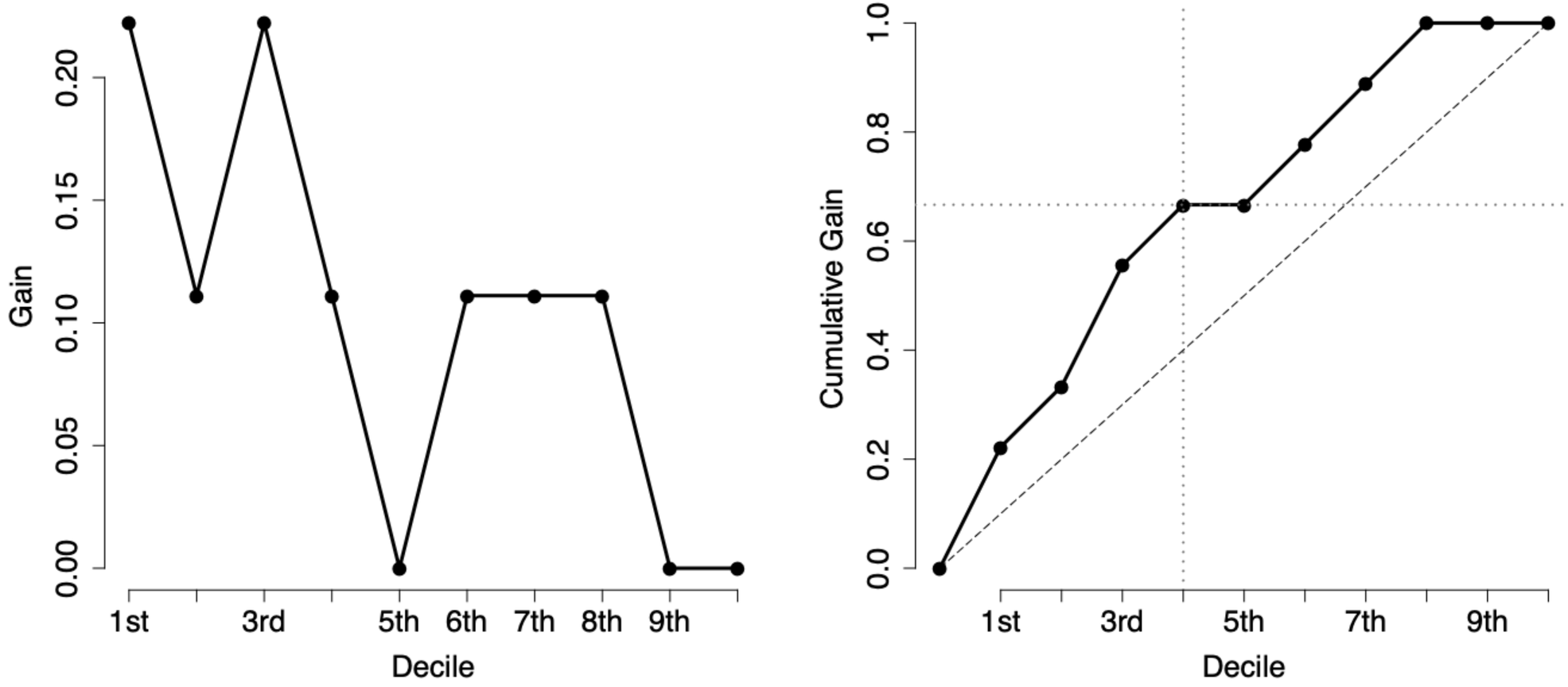
9th	15	ham	ham	0.059	TN
	13	ham	ham	0.064	TN
10th	7	ham	ham	0.001	TN
	11	ham	ham	0.003	TN

Decile	Positive (*spam*) Count	Negative (*ham*) Count	Gain	Cum. Gain
1st	2	0	0.222	0.222
2nd	1	1	0.111	0.333
3rd	2	0	0.222	0.556
4th	1	1	0.111	0.667
5th	0	2	0.000	0.667
6th	1	1	0.111	0.778
7th	1	1	0.111	0.889
8th	1	1	0.111	1.000
9th	0	2	0.000	1.000
10th	0	2	0.000	1.000

Gain and Cumulative Gain Visualized



The gain in a particular decile can be interpreted as a measure of how much better than random guessing the predictions made by a model are.

Lift of Prediction Scores

Lift

A measure of how much higher the actual percentage of positive instances in a decile ***dec*** is than the rate expected.

$$\text{lift}(\boldsymbol{dec}) = \frac{\%\text{ of positive test instances in decile } \boldsymbol{dec}}{\%\text{ of positve test instances}}$$

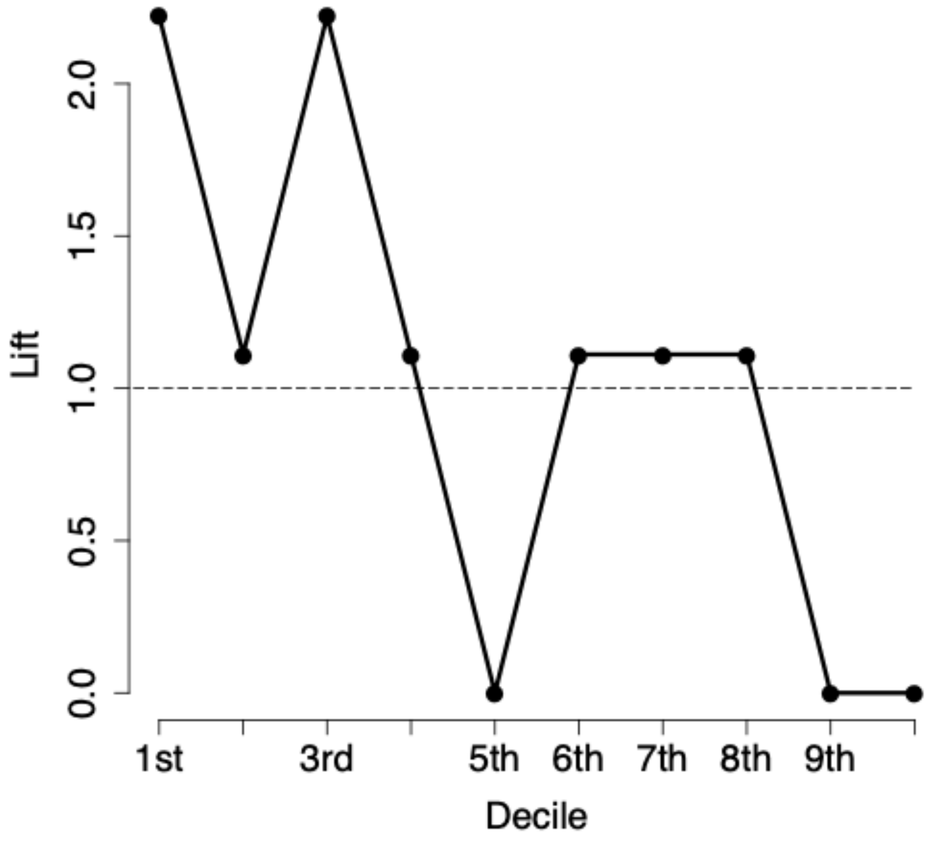
Cumulative Lift

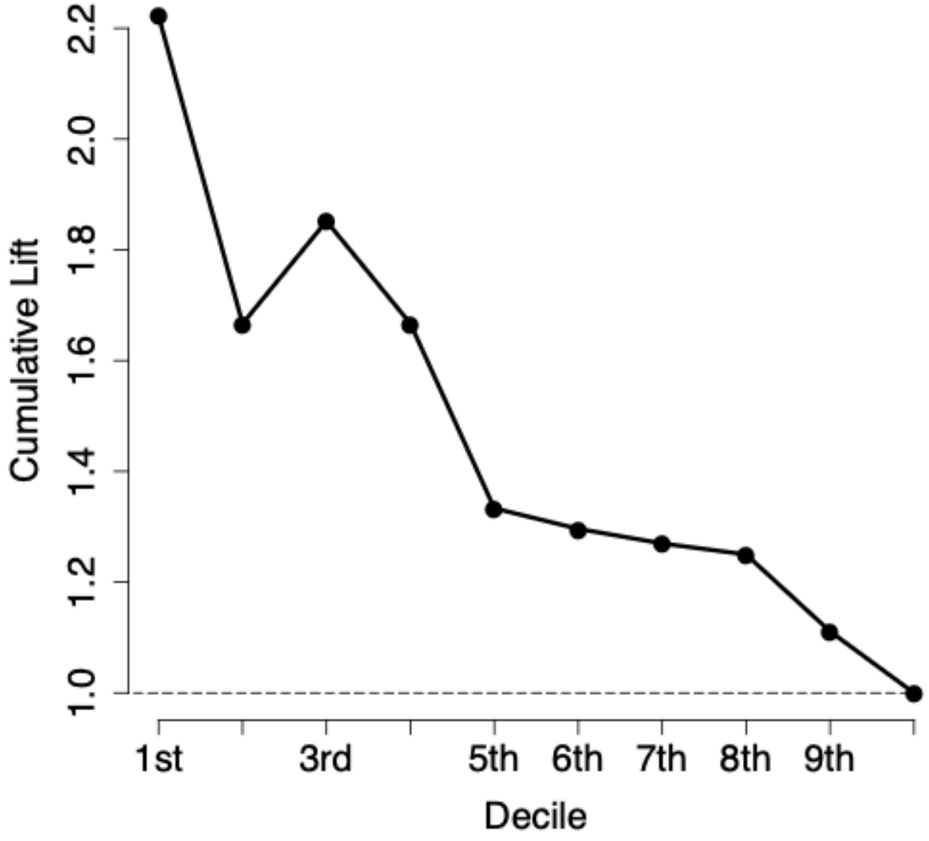
$$\text{cumulative lift}(\boldsymbol{dec}) = \frac{\%\text{ of positive test instances in all decile up to } \boldsymbol{dec}}{\%\text{ of positve test instances}}$$

Gain and Lift All Together for Email Example

Decile	Positive (*spam*) Count	Negative (*ham*) Count	Gain	Cum. Gain	Lift	Cum. Lift
1st	2	0	0.222	0.222	2.222	2.222
2nd	1	1	0.111	0.333	1.111	1.667
3rd	2	0	0.222	0.556	2.222	1.852
4th	1	1	0.111	0.667	1.111	1.667
5th	0	2	0.000	0.667	0.000	1.333
6th	1	1	0.111	0.778	1.111	1.296
7th	1	1	0.111	0.889	1.111	1.270
8th	1	1	0.111	1.000	1.111	1.250
9th	0	2	0.000	1.000	0.000	1.111
10th	0	2	0.000	1.000	0.000	1.000

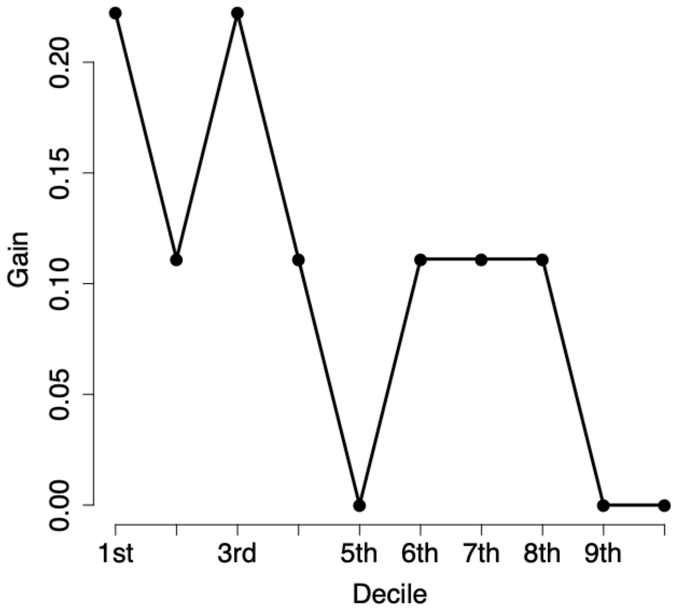
Lift and Cumulative Lift Visualized

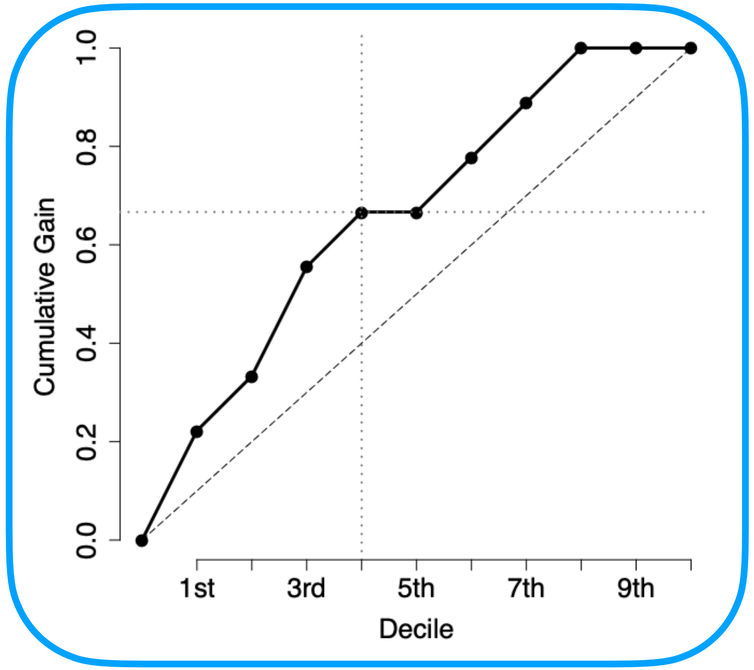


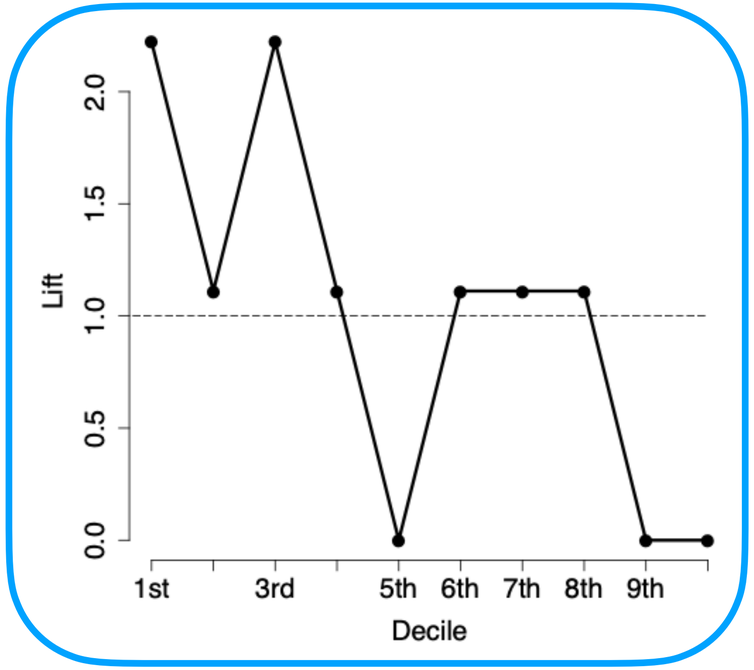


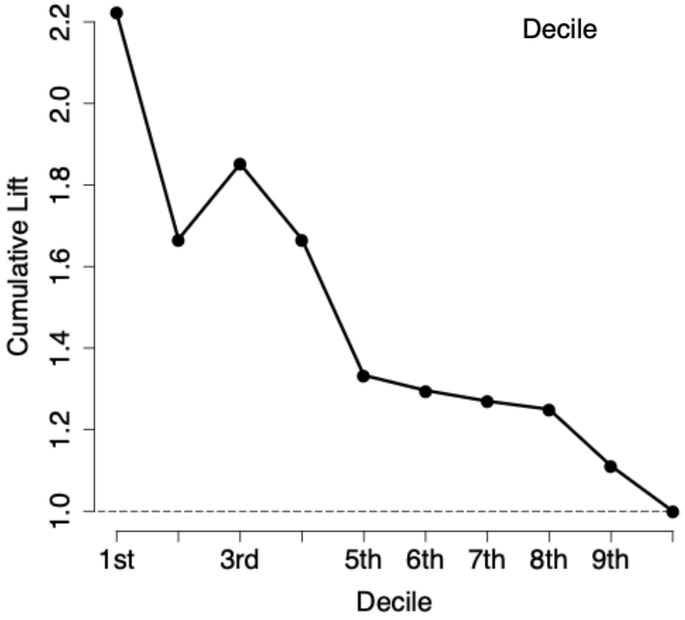
For a well-performing model, the lift curve should start well above 1.0 and cross 1.0 at one of the lower deciles. Lift range is $[0,\infty]$.

Gain and Lift Visualized On One Slide

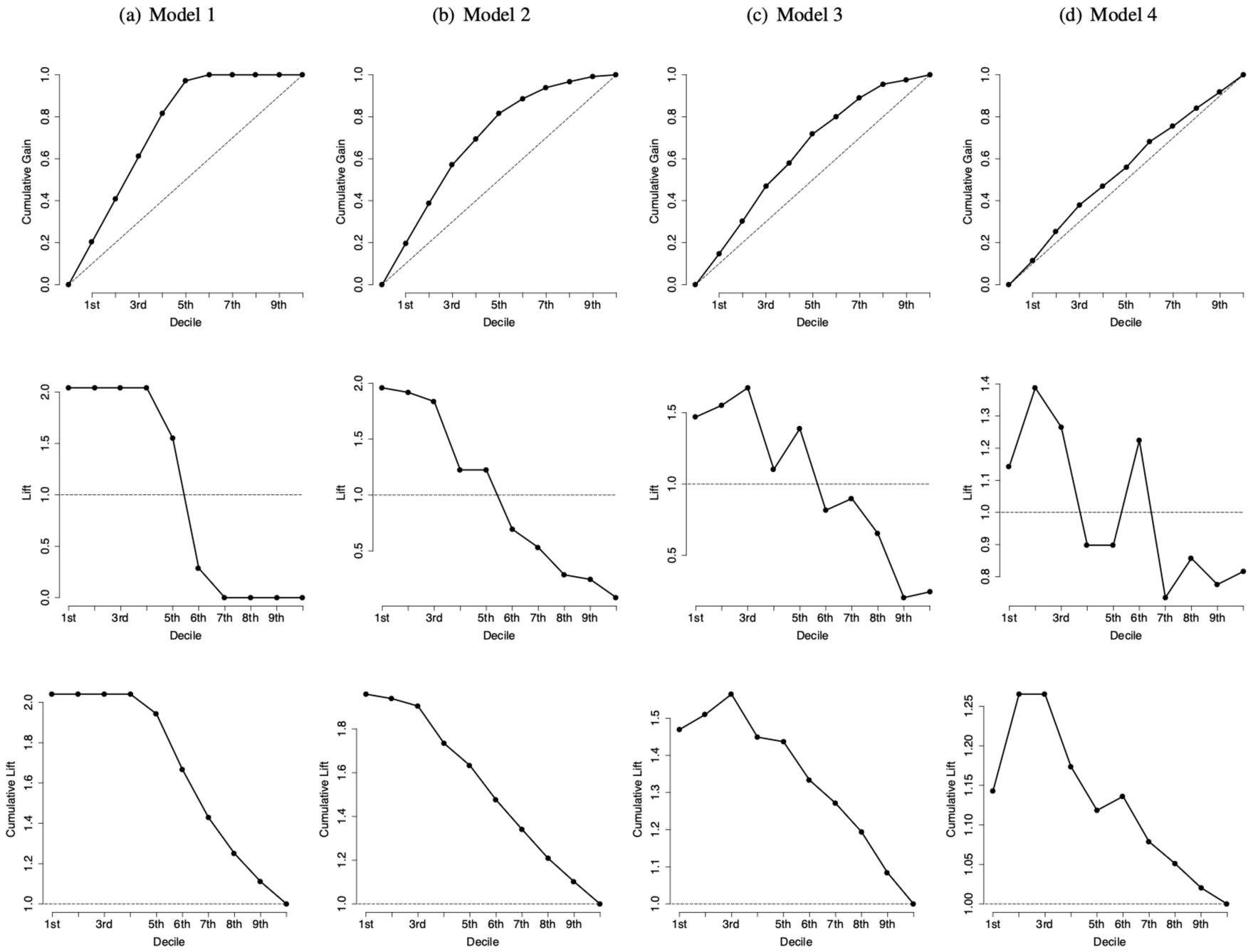






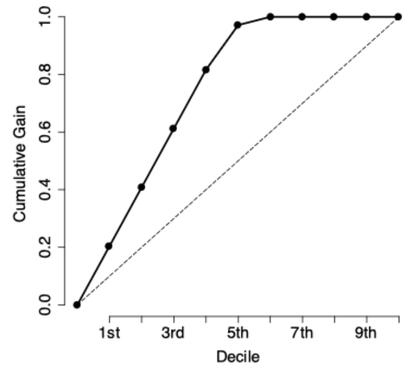


Comparison Models using Cumulative Gain, Lift, and Cumulative Lift

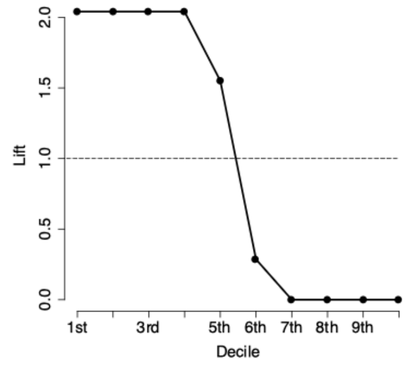


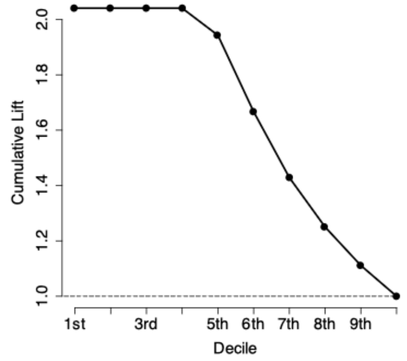
Lets Write Down Some Observations about Model 1

(a) Model 1



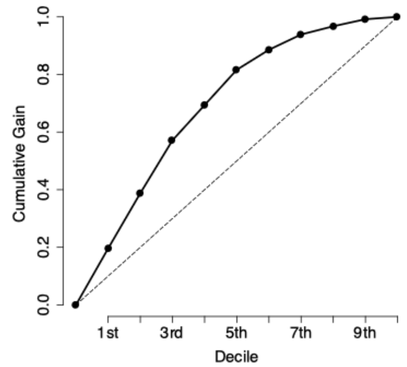
$80\,\%$ of the spam messages are identified in the top $40\,\%$ of the model predictions.

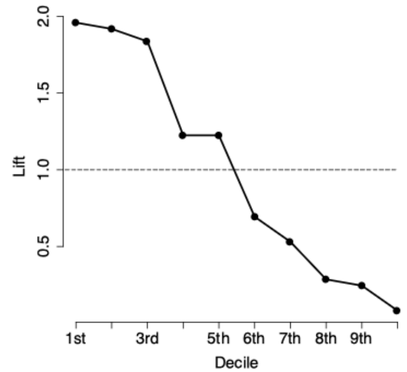


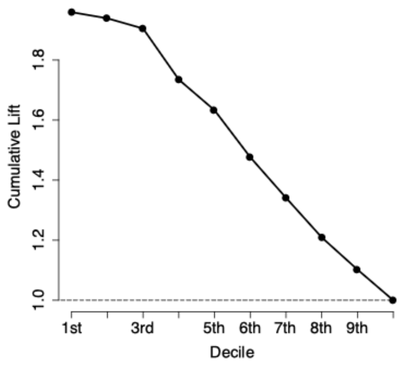


Lets Write Down Some Observations about Model 2

(b) Model 2



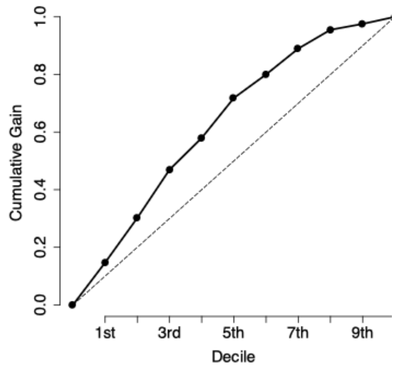




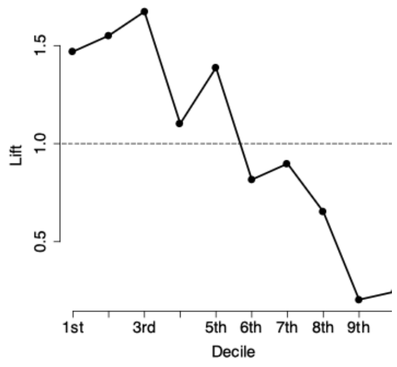
$80\,\%$ of the spam messages are identified in the top $50\,\%$ of the model predictions.

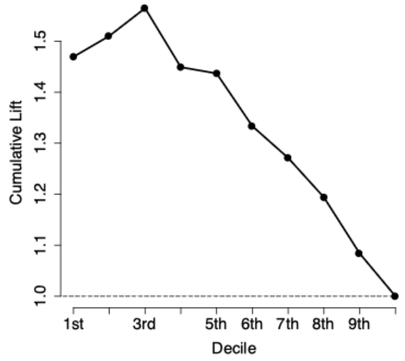
Lets Write Down Some Observations about Model 3

(c) Model 3



$80\,\%$ of the spam messages are identified in almost the top $60\,\%$ of the model predictions.





What is the "Ideal" Shape of Cumulative Gain, Lift and Cumulative Gain?

We need there brave volunteers!

This Lecture Covers ...

- Prediction Scores (two-class)
  - ROC curve.
  - Area under the curve (ROC index).
  - Gain and lift.

- Performance Measure
  - Multinomial targets.
  - Continuous targets.

— *ML for Predictive Data Analytics*

Multinomial Targets

Up until now, we have considered a two level (two class) problem.
However, there are many prediction problems that are multinomial.

Problems with multiple target levels (values).

We now discuss measures and approaches to evaluate them.

Confusion Matrix for Multinominal Targets

		Prediction					
		level 1	*level 2*	*level 3*	$\cdots$	*level l*	Recall
Target	*level 1*	-	-	-		-	-
	level 2	-	-	-		-	-
	level 3	-	-	-		-	-
	$\vdots$				$\ddots$		$\vdots$
	level l	-	-	-		-	-
	Precision	-	-	-	$\cdots$	-	

TP(2)

FP(2)

TP(l): # of instances correctly predicted as l.

FP(l): # of instances incorrectly predicted as l.

FN(l): # of instances that should have been predicted as l.

figure from ML for Predictive Data Analytics textbook.

Precision and Recall for Every Class in Multinomial Targets

Precision for class l

$$\text{precision}(l) = \frac{\text{TP}(l)}{\text{TP}(l) + \text{FP}(l)}$$

Recall for class l

$$\text{recall}(l) = \frac{\text{TP}(l)}{\text{TP}(l) + \text{FN}(l)}$$

Example of Confusion Matrix on Bacterial Species Identification

		Prediction				
		durionis	*ficulneus*	*fructosus*	*pseudo.*	Recall
Target	*durionis*	5	0	2	0	0.714
	ficulneus	0	6	1	0	0.857
	fructosus	0	1	10	0	0.909
	pseudo.	0	0	2	3	0.600
	Precision	1.000	0.857	0.667	1.000	

A good model has high values on the diagonal

Lets calculate a few metrics.

Classification Accuracy and Recall

$$\text{ACC} = \frac{(5+6+10+3)}{(5+6+10+3+1+2+1+2)} = 80\,\%$$

		Prediction				
		durionis	*ficulneus*	*fructosus*	*pseudo.*	Recall
Target	*durionis*	5	0	2	0	0.714
	ficulneus	0	6	1	0	0.857
	fructosus	0	1	10	0	0.909
	pseudo.	0	0	2	3	0.600
	Precision	1.000	0.857	0.667	1.000	

However, accuracy for each class (Recall) is

C1: 0.714, C2: 0.857, C3: 0.909, and C4: 0.6

Hence, average class accuracy is lower than the overall accuracy (useful to compute when evaluating the model performance). Recall and ACA are particularly important to highlight if the model has very poor performance for one class.

figure from ML for Predictive Data Analytics textbook.

Average Classification Accuracy (ACA)

		Prediction				
		durionis	*ficulneus*	*fructosus*	*pseudo.*	Recall
Target	*durionis*	5	0	2	0	0.714
	ficulneus	0	6	1	0	0.857
	fructosus	0	1	10	0	0.909
	pseudo.	0	0	2	3	0.600
	Precision	1.000	0.857	0.667	1.000	

$$\text{Average Class Accuracy}_{HM} = \frac{1}{\frac{1}{|\text{levels}(t)|} * \sum_{k \in \text{levels}(t)} \frac{1}{\text{recall}_k}}$$

$$\text{ACA}_{HM} = \frac{1}{\frac{1}{4} * \left(\frac{1}{0.714} + \frac{1}{0.857} + \frac{1}{0.909} + \frac{1}{0.600} \right)} = \frac{1}{1.333} = 75\,\%$$

figure from ML for Predictive Data Analytics textbook.

F_1 Measure for Every Class in Multinomial Targets

F_1-Score

$$F_1 \text{ score}(l) = \frac{2 * \text{precision}(l) * \text{recall}(l)}{\text{precision}(l) + \text{recall}(l)}$$

$$F_1 \text{ score}(l) = \frac{\text{TP}(l)}{\text{TP}(l) + \frac{1}{2} * (\text{FP}(l) + \text{FN}(l))}$$

How to combine $F_1(l)$ or $\text{TP}(l)$, ... to get an overall understanding of F_1 to multinomial problems? (next slides)

Macro and Micro F_1 Scores for Multinomial Targets Problem

Aggregation of $F_1(l)$ calculated individually based on $TP(l)$, $FP(l)$, and $FN(l)$ and each class.

Macro F_1 Score

$$\text{Macro } F_1 = \frac{\sum_l F_1(l)}{\#\text{ of classes}}$$

Micro F_1 Score

$$\text{Micro } F_1 = \frac{TP}{TP + \frac{1}{2} * (FP + FN)}$$

Aggregation of $TP(l), \ldots$ to get a TP over all classes, then compute the score.

Intuitions on Macro and Micro F_1 Scores

Macro F_1 Score

It gives each class equal importance.

$$\text{Macro } F_1 = \frac{\sum_l F_1(l)}{\#\text{ of classes}}$$

Micro F_1 Score

It gives equal importance to each observation.

$$\text{Micro } F_1 = \frac{TP}{TP + \frac{1}{2} * (FP + FN)}$$

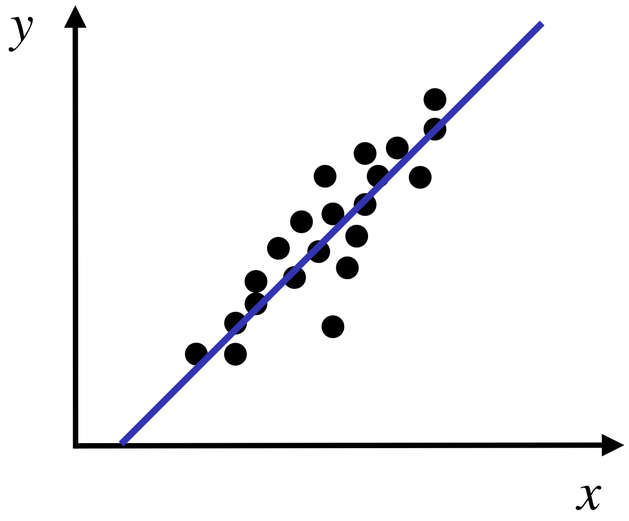
Micro F_1 score performs worse on **imbalanced** datasets than **macro** F_1 score.

Continuous Targets

Up until now, we had considered categorical targets (i.e., yes/no, cat/dog type predictions). However, there are many prediction problems where the target is continuous.

x-axis: the set of features.

y-axis: the target value (stock price).



Basic Measures of Error for Continuous Targets

Sum of Squared Errors

$$\text{sum of squared errors} = \frac{1}{2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where $\hat{y}_i = f_\theta(\boldsymbol{x}_i)$ is the model prediction.

n: number of examples in the dataset.
i: denote the example i.
θ: the set of model parameters.
$\boldsymbol{x}$: the input features.
$f(\cdot)$: is the model which maps your input $\boldsymbol{x}$ to the estimate of the target y.

Sum of Squared Errors Can Take Other Variations

Range $[0,\infty]$. Smaller, the better

Mean Squared Error (MSE)

$$\text{mean squared error} = \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}$$

Root Mean Squared Error (RMSE)

$$\text{root mean squared error} = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}}$$

Intuition on the Unit of Root Mean Squared Error

RMSE values are in the same units as the target value; hence, the values are more meaningful about what the error for predictions are.

Ex. Consider a drug dosage prediction problem where the target value have unit of mg. For an RMSE of 1.380, we can expect the prediction made by the regression model to be $1.38\ mg$ out on average.

Note: due to the squared term (we discussed in previous lectures), RMSE tends to overestimate error slightly as it overemphasizes individual large errors.

Sum of Squared Errors Can Take Other Variations

Mean Absolute Error (MAE)

Range $[0,\infty]$. *Smaller, the better*

$$\text{mean absolute error} = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{n}$$

MAE values are in the *same* units as the target value; hence, the values are more easily meaningful about what the error for predictions are (see example in previous slide).

Suggested (more common) to use RMSE over MAE as it is better to be pessimistic when estimating the performance of models.

How small an MSE, MAE, or RMSE error is good?

1 Go to wooclap.com

2 Enter the event code in the top banner

Event code
QUGLUJ

Enable answers by SMS

How Small An MSE, MAE, or RMSE Error is Good?

MSE, MAE, RMSE values are domain dependent. Hard to judge without deep knowledge of a domain! Hence, we introduce a new domain independent measure of model performance.

R^2 Coefficient

Range $[0,1)$. *Larger, the better*

$$R^2 = 1 - \frac{\text{sum of squared errors}}{\text{total sum of squares}}$$

$$\text{sum of squared errors} = \frac{1}{2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \qquad \text{total sum of squares} = \frac{1}{2}\sum_{i=1}^{n}(y_i - \bar{y})^2$$

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

What R^2 is doing? Lets Interpret It

$$R^2 = 1 - \frac{\text{root mean squared error}}{\text{total sum of squares}}$$

Root mean squared error: Unexplained variation in the data.

Total sum of squares: Total variation in the data.

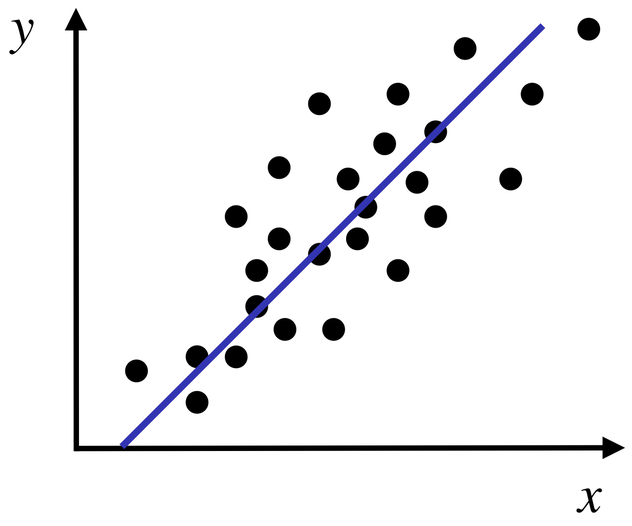
R^2 quantifies the amount of variations in the target feature that is explained by the descriptive features in the model.

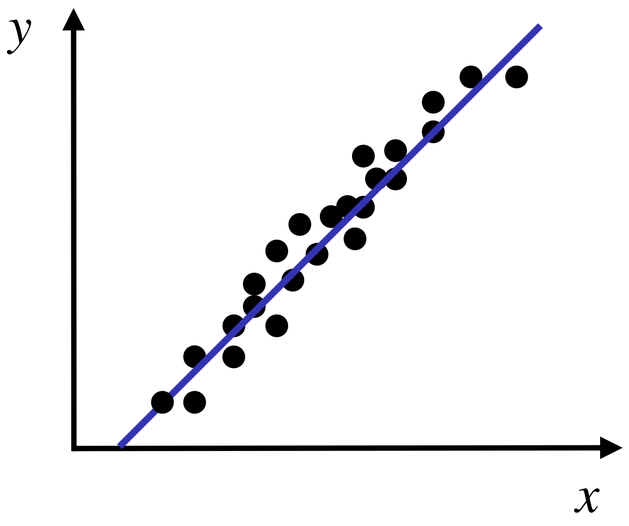
Question. Consider a imaginary model that always predicts the target to be the average value of the test set. What would be the R^2?

Intuitively, that model does not take into account the variation in the data; hence, it should give you a bad R^2. Indeed, the R^2 in this case is 0.

Taking a Closer Look at R^2

Which one of the models have higher R-squared?





Takeaways

- ***Performance measure for categorical target*** — *prediction scores.*

- ***Receiver Operating Characteristic Curve (ROC)***
  — *True positive rate vs. False positive rate.*

- ***Gain and lift*** — *Focus on one positive target value.*

- ***Multinomial target performance measure*** — *extension of confusion matrix.*

- ***Continuous target performance measure*** — *square/absolute errors,* R^2.